

4. Sequences and series		
4.1	Sequences, including those given by a formula for the n th term and those generated by a simple relation of the form $x_{n+1} = f(x_n)$.	
4.2	Understand and work with arithmetic sequences and series, including the formula for the n th term and the sum of a finite arithmetic series; the sum of the first n natural numbers.	The proof of the sum formula should be known. Understanding of $\sum$ notation will expected.
4.3	Increasing sequences, decreasing sequences and periodic sequences.	
4.4	Understand and work with geometric sequences and series, including the formulae for the n th term and the sum of a finite geometric series; the sum to infinity of a convergent geometric series, including the use of $ r < 1$.	For example, given the sum of a series students should be able to use logs to find the value of n. The proof of the sum formula for a finite series should be known. The sum to infinity may be expressed as S_∞.
4.5	Binomial expansion of $(a + bx)^n$ for positive integer n .	The notations $n!$, $\binom{n}{r}$ and nC_r may be used.